

Eletrostatics

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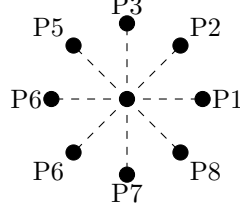
Contents

1 Punctual loads

3

1 Punctual loads

A punctual load in space will create an electric field towards all surrounding points in space.



In order to measure a punctual load a reference point is needed. For instance, the electric field generated by the punctual load in the center, point $P0$, on point $P1$ is:

$$\vec{E}_{P0 \rightarrow P1} = k_e \frac{p_0}{r^2} \hat{r}$$

where p_0 is the electric charge at point $P0$ in Coulombs (C), r the distance between point $P0$ and the reference point $P1$ in meters (m), $\hat{r}$ is the unit vector indicating the direction of an electric field, and k_e is the Coulomb's constant

$$k_e = \frac{1}{4\pi\epsilon_0} \approx 9 \cdot 10^9$$

and ϵ_0 is the Vacuum permittivity.

A punctual load also produces a force to all surrounding points in space. The force of an electric field is written as

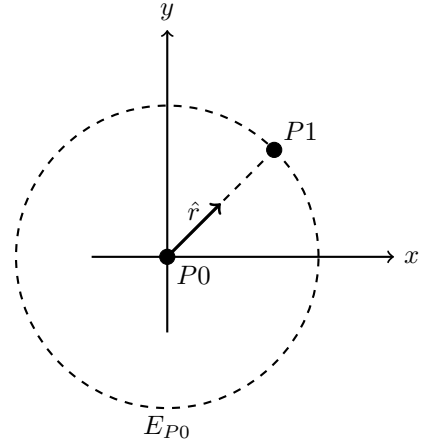
$$\begin{aligned} \vec{F}_{P0 \rightarrow P1} &= p_1 \vec{E}_{P0 \rightarrow P1} \\ &= \frac{1}{4\pi\epsilon_0} \frac{p_0 p_1}{r^2} \hat{r} \end{aligned}$$

The unit of an electric force is Newtons (N) and the unit of an electric charge is Coulombs (C). Therefore, the unit of an electric field is Newtons/Coulombs ($\frac{N}{C}$).

$$\begin{aligned} F &= \frac{1}{4\pi\epsilon_0} \frac{qq'}{r^2} \implies \frac{C^2}{m^2} = N \\ E &= \frac{F}{q} \implies \frac{C^2}{m^2} \frac{1}{C} = \frac{N}{C} \end{aligned}$$

A punctual load produces an electric potential to all surrounding points in space. The unit of electric potential is Volts (V).

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r} \implies \frac{C}{m} = V$$



Vectors associated to a punctual load with *positive* charge will point outwards. Vectors associated to a punctual load with *negative* charge will point inwards.

